ARUNANGSHU PRAMANIK

+91-78906-69228 ◇ Bangalore (Bengaluru), Karnataka, India
arunangshu01@gmail.com
LinkedIn ◇ GitHub ◇ LeetCode ◇ Portfolio

SUMMARY

Senior Software Engineer with 7+ years of experience architecting data and cloud solutions across Azure and GCP. Proven track record of designing and delivering production-grade data pipelines, SaaS platforms, and AI-powered tooling - spanning BigQuery and Airflow ingestion frameworks, near real-time integration architectures, intelligent agents, and data-catalog assistants for Enterprise Retail & FMCG customers.

EDUCATION

Bachelor of Technology in Computer Science & Engineering 2015-2019
Institute of Engineering & Management, Kolkata DGPA: 8.53

SKILLS

Programming Languages:	Python, SQL, Java, JavaScript, TypeScript
Databases & Data Warehouses:	Microsoft SQL Server, Oracle, PostgreSQL, Snowflake, Databricks, BigQuery
Data Analysis:	Pandas, NumPy, PySpark
Frameworks:	FastAPI, Django, Spring Boot, RESTful APIs
Cloud Platforms:	Azure, GCP
Orchestrator:	Apache Airflow
Azure Resources:	Azure Event Hubs, Azure Files, Azure Blob Storage, Azure Storage Explorer, Azure Container Registry, Azure Container Services, Azure Key Vault
GCP Resources:	Cloud Tasks, Cloud Run, Cloud Run Functions, Cloud SQL, Cloud Scheduler, Workflows, Eventarc, Cloud Storage, Secret Manager, Pub/Sub, Dataflow, Compute Engine, Analytics Hub, IAM, VPC, Cloud Networking, Google Cloud APIs, Databricks Workspace
Devops:	Docker, Kubernetes, Jenkins, Git, GitHub, GitHub Actions, Bitbucket, Bitbucket Pipelines, Shell Scripting, YAML
Monitoring & Logging:	Datadog, Elasticsearch, Kibana
Messaging & Caching:	Kafka, Redis
Collaboration & Project Management:	Jira, Confluence, Slack, Microsoft Teams
Database & API Tools:	DBeaver, Offset Explorer, Postman
IDEs & Editors:	Visual Studio Code, PyCharm, IntelliJ IDEA Community Edition, Eclipse IDE, Python IDLE
AI Tools:	Cursor, Devin AI
Web Technologies:	HTML, CSS, Bootstrap, Angular, React

EXPERIENCE

Senior Software Engineer December 2024 - Present
Impact Analytics *Bangalore (Bengaluru), Karnataka, India*
Clients / Customers: Retail & FMCG Enterprises - Notable Enterprises include Bealls, Ralph Lauren, Peter Millar, Swarovski, Gap, Foot Locker, Patagonia, Under Armour, and Levi's

- Design, develop, test, and integrate core data engineering pipelines powering the organization's AI-driven retail product suite - PlanSmart, DemandSmart, InventorySmart, PriceSmart, PromoSmart, MarkSmart, and DataSmart - for Retail and FMCG customers.

- Architected a two-stage Airflow and GCP pipeline: a Sourcing layer that extracts data from heterogeneous databases, warehouses, cloud storage, and file transfer systems into BigQuery, and an Ingestion layer that cleanses, validates, and transforms the data into master entity tables consumed by in-house products.
- Integrated client data across OLTP and OLAP databases, data warehouses and workspaces, cloud storage, and file transfer connectors for every retailer subscribed to the product suite.
- Architected the pipeline codebase and designed a near real-time integration framework covering ingestion and cancellation flows, leveraging Cloud SQL, Cloud Run Functions, Cloud Tasks, BigQuery, Cloud Storage, Secret Manager, Workflows, Eventarc, and PostgreSQL.
- Collaborated with cross-functional team members in design sessions to define pipeline architecture, data models, and integration strategies.
- Built an AI agent for internal pipelines that detects and diagnoses failed database, warehouse, cloud storage, and file transfer connectors based on pipeline flow, significantly reducing manual triage effort.
- Developed a chatbot that generates pipeline tasks and processes tailored to client requirements and data-driven insights.
- Developed an AI chatbot and search engine spanning databases, warehouses, workspaces, and BigQuery - comparable to the native assistants in Databricks and Snowflake - offering a browsable data catalog, schema-based data inspection, table relationship mapping, and SQL query generation from natural-language requirements.
- Built cost and runtime reporting dashboards for Cloud Run, Cloud Run Functions, and Compute Engine VMs, providing granular visibility into cloud spend and resource utilization.
- Integrated Google Authentication SSO for Airflow services, enabling secure, seamless user login through the Airflow UI.
- Lead and mentor a team of junior engineers, developing their technical capabilities and reviewing their deliverables.
- Upgraded standalone codebases and legacy systems, improving code efficiency and accuracy by 90%.
- Diagnosed and resolved customer issues, identifying recurring bottlenecks on both internal and client sides to prevent repeat incidents.

Digital Engineering Lead Engineer August 2021 - December 2024
Apisero Inc. (Now Acquired by NTT Data Services) *Bangalore (Bengaluru), Karnataka, India*
Clients / Customers: Blue Yonder

- Developed a SaaS application built on event-driven architecture using Azure Event Hubs, ELK Stack, Snowflake, Azure Blob Storage, and Python.
- Modernized and modularized legacy codebases to contemporary development standards within a Data Platform, delivering real-time data pipeline solutions to over a dozen enterprise customers, including JCPenney, ASDA, 7-Eleven, CVS, Heineken, and Walmart.
- Designed and maintained ETL applications for analytics platforms, processing billions of transaction records across the customer base.
- Identified and resolved customer-facing bottlenecks, improving the integration and utilization of OLAP and OLTP databases across homogeneous and heterogeneous data migration processes.
- Managed and upgraded standalone codebases and existing systems, achieving over 90% code coverage while deploying software tools, processes, visualization, and metrics.
- Partnered with senior stakeholders to translate business requirements into configuration blueprints for data migration platforms, enhancing customer adoption and analytics capabilities.

Systems Engineer October 2019 - July 2021
Infosys Limited *Bangalore (Bengaluru), Karnataka, India*
Clients / Customers: Cisco Systems

- Researched, designed, implemented, and managed software programs to ensure optimal performance and scalability.
- Collaborated with project team members to identify and deliver modifications and enhancements to existing programs.
- Authored efficient, maintainable code with a focus on operational practicality and robustness.

- Developed client-side and server-side architecture using modern front-end and back-end technologies.
- Engineered and maintained robust databases and applications supporting core business needs.
- Designed and documented APIs, rigorously testing features to ensure responsiveness and efficiency.
- Implemented comprehensive security and data protection measures.
- Tested and evaluated new programs, establishing quality assurance procedures that achieved 95% efficiency.
- Maintained and upgraded existing systems, deploying software tools, processes, and metrics to improve performance.

PERSONAL PROJECTS

M.A.H.A.K.A.L. October 2026 - Present

Description: *M.A.H.A.K.A.L.: Modular Agentic Hub for Automation, Knowledge & Adaptive Logic. A personal voice AI assistant built with FastAPI and React, powered by Groq (GPT-OSS 120B + Whisper) with Gemini fallback, plus Edge TTS.* **(Live Link: Try it here)**

Technologies Used: Python, FastAPI, Groq, Whisper, Gemini, Edge TTS, React, Vite, Firestore, Cloud Run, Firebase Hosting, Docker

- Voice and text conversations with streamed responses, Whisper speech recognition, and Edge TTS playback.
- Automatic Gemini fallback when the primary Groq provider fails or reaches its rate limit.
- Google sign-in with guest-to-user conversation migration, persistent Firestore history, and long-term memory.
- Tool calling for weather, time, Wikipedia lookup, and remembering or forgetting user preferences.
- Cloud Run and Firebase Hosting deployment with Secret Manager, keyless GitHub authentication, and CI-gated releases.

doppel - Mock API playground November 2026 - Present

Description: *doppel: your backend stand-in. A schema-driven mock REST API generator that turns a JSON schema into a working API with realistic seeded data, resource relationships, and an interactive playground. Built with FastAPI, React, and Faker.* **(Live Link: Try it here)**

Technologies Used: Python, FastAPI, Faker, React, Vite, Cloud Run, Firebase Hosting, Docker, GitHub Actions

- Define resources, field types, and ref() foreign-key links in JSON, then generate reproducible data from a fixed seed.
- Typed generators for uuid, person, email, date, datetime, money, enums, sentences, and more across five locales.
- Dynamic REST endpoints with pagination, exact-match filtering, sorting, and per-dataset OpenAPI docs.
- Interactive playground with Table and JSON views, cURL copying, starter templates, and schema import/export.
- Isolated read-only public demo and per-user playgrounds on Cloud Run and Firebase Hosting, with bounded request simulation and CI-gated keyless deploys.

Flowbench - data pipeline playground November 2026 - Present

Description: *Flowbench: build a pipeline, break it, inspect every row. An interactive data pipeline playground that runs real validation and transformation pipelines on seeded datasets, compiles them to DuckDB SQL, and lets you open every stage: records in and out, rejections with reasons, and record-level lineage. Built with FastAPI and React.* **(Live Link: Try it here)**

Technologies Used: Python, FastAPI, DuckDB, React, Vite, Cloud Run, Firebase Hosting, GitHub Actions

- Typed pipeline builder: every validator and transform has a schema-aware editor with field pickers limited to the fields that exist at that stage.
- Static planning before execution: the planner propagates the schema through each stage and reports errors and warnings; the API refuses to run a pipeline that has errors.
- Real, bounded execution: rows stream lazily through stage iterators with exact, conserved counts (in = out + rejected + filtered) and per-stage timing.
- Stage-level inspection: sampled input, output, and rejected records, a per-field profile over all rows, and schema in vs. out for every node.
- Record-level lineage: every source row carries an ID you can trace through the stages, reporting honestly when a trace is not possible.

- Failure to fix to replay to compare: a deliberately faulty dataset demonstrates rejection handling, replay-with-edits that links back to the original run, and run comparison that tolerates added or removed stages.

AWARDS & RECOGNITIONS

- Cisco Systems: Most Valuable Player of the Quarter – Infosys Limited (March 2020)
- Rookie of the Quarter – Apisero Inc. (October 2021)
- Apisero Aces Award – Apisero Inc. (October 2022)
- 5 x Raving Reviews Awards – Apisero Inc. (October 2021, December 2021, March 2022, January 2023 & February 2023)
- Vendustrious Award – Apisero Inc. (March 2023)
- Impact Excellence Award: Q2 2026 - Impact Analytics (June 2026)